

Presentation II: Research / Contribution

DiT + Diffusion Forcing for Interactive Billiards Physics

Your Name

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Agenda

- ➊ Research question and novelty
- ➋ What we built (data $\rightarrow$ VAE $\rightarrow$ world model $\rightarrow$ interactive inference)
- ➌ Quantitative findings (historical and near-fair reruns)
- ➍ Qualitative findings + failure modes
- ➎ Hardships/debugging journey
- ➏ Final takeaways and next steps

Research Question and Hypothesis

Research question:

Can diffusion forcing help an action-conditioned DiT preserve interactive billiards physics better than baseline single-step training?

Hypothesis:

- DF should improve long-horizon rollouts by training on partially self-generated context.
- Expected gains: collision continuity and wall-bounce stability.
- Remaining risk: identity consistency (color drift/swallowing artifacts).

What Is Novel Here?

- Focus on **interactive physics consistency**, not only image quality.
- Controlled multi-dataset strategy (v1/v2/v3) to stress different behaviors.
- Direct baseline-vs-DF rollout comparisons with fixed horizons.
- Explicit failure analysis: collision validity vs identity consistency.

Project Journey (What We Actually Did)

- Built large simulation pipeline and collected staged datasets (v1/v2/v3).
- Trained VAE and exported latent shards for scalable world-model training.
- Trained DiT baselines, then implemented and trained Diffusion Forcing (DF).
- Scaled to larger runs (joint datasets, up to $\sim 1.52\text{B}$ model line).
- Built interactive inference/UI routes and tested action-conditioned behavior.

Main scientific thread: does DF preserve rollout physics better?

Dataset Strategy (v1/v2/v3)

Dataset	Episodes	Frames/ep	Purpose
v1	100k	600	Baseline coverage
v2	100k	600	More diversity/randomization
v3	200k	600	Collision-heavy hard cases

This staged design improved robustness and made collision behavior easier to learn.

World Model Setup We Compared

- Same backbone family for key baseline/DF comparisons:
 - $d_{\text{model}} = 768$, 10 layers, 12 heads, context length $L = 8$
- Baseline objective: single-step diffusion noise prediction
- DF objective: rollout training with context updates under scheduled teacher forcing
- Rollout eval: horizons $h = \{1, 4, 8, 16, 32\}$, DDIM=10, same VAE decoder

Experimental Setup (Main Comparison)

Historical comparison (reported earlier):

- Baseline: 60M samples
- DF: 120M total samples

Near-fair check (run now):

- Baseline exact: 60.0M checkpoint
- DF smoke: 60.2M checkpoint (resume + 200k with DF)
- Same eval pipeline, same horizons, same DDIM steps

Historical Result (60M Baseline vs 120M DF)

Model	MSE@16	MSE@32	PSNR@16	PSNR@32
Baseline 60M	54.023	318.312	17.133	14.857
DF 120M	0.763	0.827	19.447	19.372

Caveat: this is strong but not a strict apples-to-apples budget comparison.

Key Result Table (Near-Fair 60.0M vs 60.2M)

Model	MSE@16	MSE@32	PSNR@16	PSNR@32
Baseline 60.0M	54.023	318.312	17.133	14.857
DF 60.2M	1.238	1.597	18.809	18.063

Source: rollout_eval_ctx8_baseline60m_vs_df60p2m_20260303

Sanity Check (Different Eval Seed)

- Repeated eval with a different seed (different sampled clips).
- Pattern stayed the same:
 - Baseline MSE@16/32 remained very high.
 - DF MSE@16/32 remained low.

Metric	Baseline (seed123)	DF (seed123)
MSE@16	105.241	1.268
MSE@32	414.579	1.378

Control Check: Is It Just +200k Training Steps?

- We ran a baseline control from 60.0M $\rightarrow$ 60.2M (no DF) with matched resume-smoke style.
- Result: baseline control got worse at long horizon.

Model	MSE@16	MSE@32
Baseline 60.0M	54.0	318.3
Baseline control 60.2M	2022.5	5379.9
DF 60.2M	1.289	1.766

Interpretation: Why This Big Gap Can Happen

- Inference is autoregressive: model consumes its own predicted context.
- One-step baseline is underexposed to this regime during training.
- DF explicitly trains multi-step context replacement:

$$\tilde{z}_{k+1} = m_k z_{k+1}^{gt} + (1 - m_k) \hat{x}_{0,k}$$

- This reduces train/inference mismatch and mitigates long-horizon error explosion.

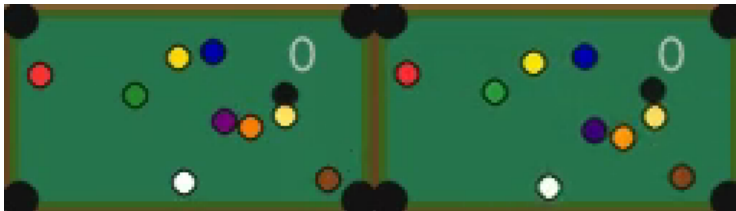
Qualitative Findings

What works better

- wall bounces generally correct
- collisions more coherent
- less long-horizon drift

What still fails

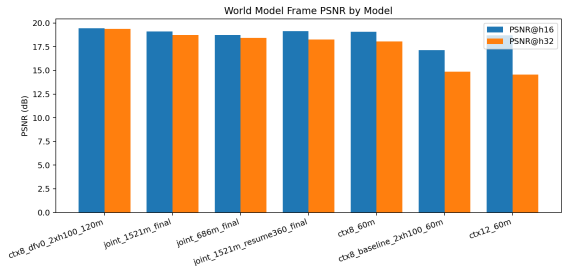
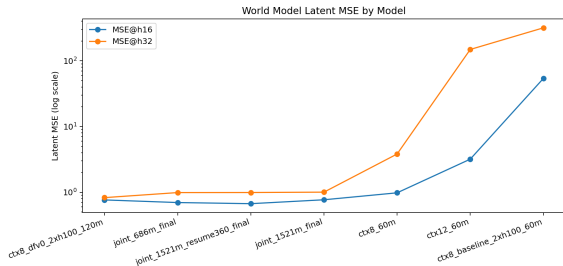
- color identity drift
- occasional swallowing/merging
- interactive latency still high



Scaling Findings (Beyond Baseline vs DF)

- As we scaled data/model, short-to-mid horizon quality generally improved.
- Gains were not strictly monotonic for every horizon/metric.
- Practical takeaway:
 - scaling helps,
 - but training regime and rollout robustness matter just as much.

Metric Trend Visuals



These charts summarize long-horizon behavior across milestone checkpoints.

Hardships We Hit (And Why They Matter)

- **Physics artifacts:** color drift and occasional swallowing/merging.
- **Interactive inference:** high latency at low DDIM steps vs quality tradeoff.
- **Action loop:** early versions had weak action effect / unclear feedback, needed UI instrumentation.
- **Training ops:** run failures (NCCL watchdog abort, bad module path restart).
- **Evaluation rigor:** needed near-fair reruns and controls to avoid overclaiming.

Limitations and Methodological Risks

- Near-fair is 60.0M vs 60.2M, not exactly equal sample count.
- Historical headline compare (60M vs 120M) mixes regime and budget effects.
- Eval currently uses independently sampled diffusion noise per model.
- Identity consistency remains the main unresolved artifact.

Mitigation plan

- deterministic/noise-shared paired eval for strict fairness
- color-consistency auxiliary loss with simulator masks
- collision-focused sampling with uniform-mix regularization

Distillation Status (Teacher vs Student, Mar 4)

Eval run: rollout_eval_joint_teacher555m_vs_distill573m_20260304_run01

Model	MSE@16	MSE@32	PSNR@16	PSNR@32
Teacher 1521M (ckpt 555M)	0.662	0.957	19.04	18.09
Student 573M (distill 30M)	953.11	3185.85	11.06	8.70

Takeaway: current student is much faster/smaller, but quality drops too much at long horizon.

Distillation is not yet ready to replace teacher for fidelity-critical demos.

Next Steps

- ① Strict fair ablation: matched 120M baseline vs matched 120M DF.
- ② Add color consistency loss on predicted x_0 .
- ③ Stabilize distillation for faster interactive inference.
- ④ Improve interactive UI: clearer action overlays + lower-latency generation.

Current status: DF already delivers a strong long-horizon robustness gain.

Final Takeaways

- We validated the core hypothesis: DF improves long-horizon physics consistency.
- Wall bounce and collision coherence are now credible in many rollouts.
- Biggest remaining research gap is identity consistency (color/object persistence).
- This is now a solid base for a faster, more playable interactive world model.

For demo:

- VAE reconstruction GIF:
`milestones_preview/final_project_media/vae_latent_compare_ep0_13s.gif`
- Latest rollout GIF:
`milestones_preview/final_project_media/latest_resume555m_clip_002.gif`

These are lightweight assets for fast playback in class.